

ECON 712: Applied Econometrics

Lecture 13: Potential Outcomes and Causality

E.g. Sorting on Unobservables

- Suppose we have data on two internationally students observationally equivalent ($i = M, K$):
 - ▶ $D_i = 1$ if student i buys insurance (*treatment* group), $D_i = 0$ if student i does not buy insurance (*control* group)
 - ▶ Y_{1i} : health outcome of student i if s/he buys insurance, Y_{0i} : health outcome of student i if s/he does not buy insurance
- A causal effect of the policy would be equivalent to $Y_{1i} - Y_{0i}$ for either $i = M$ or $i = K$
 - ▶ Using the following table, we show that a simple empirical comparison leads to imprecise estimates of the causal effect:

Outcomes and treatments for K and M		
	K	M
Potential outcome without insurance: Y_{0i}	3	5
Potential outcome with insurance: Y_{1i}	4	5
Treatment (insurance status chosen): D_i	1	0
Actual health outcome: Y_i	4	5
Treatment effect: $Y_{1i} - Y_{0i}$	1	0

Average Treatment Effect

Definition: Average Treatment Effect (ATE)

The average treatment effect is the population average of all i individual treatment effects

$$E[\delta_i] = E[Y_{1i} - Y_{0i}] = E[Y_{1i}] - E[Y_{0i}]$$

- Cannot be calculated because Y_{1i} and Y_{0i} do not exist *for the same unit*

Average Treatment Effect dgd on the Treated

Definition: Average Treatment Effect on the Treated (ATT)

The average treatment effect on the treatment group is equal to the average treatment effect conditional on being a treatment group member:

$$E[\delta_i \mid D = 1] = E[Y_{1i} - Y_{0i} \mid D = 1] = E[Y_{1i} \mid D = 1] - E[Y_{0i} \mid D = 1]$$

- Cannot be calculated because Y_{1i} and Y_{0i} do not exist *for the same unit*

Average Treatment Effect on the Untreated

Definition: Average Treatment Effect on the Untreated (ATU)

The average treatment effect on the untreated group is equal to the average treatment effect conditional on being untreated:

$$E[\delta_i \mid D = 0] = E[Y_{1i} - Y_{0i} \mid D = 0] = E[Y_{1i} \mid D = 0] - E[Y_{0i} \mid D = 0]$$

- Cannot be calculated because Y_{1i} and Y_{0i} do not exist *for the same unit*

Independence

Independence Assumption

Treatment is assigned to a population independent of that population's potential outcomes:

$$(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i$$

- This is random or quasi-random assignment and ensures mean potential outcomes for the treatment group and control group are the same. Also ensures other variables are distributed the same for a large sample:

$$\begin{aligned} E[Y_{0i} \mid D = 1] &= E[Y_{0i} \mid D = 0] \\ E[Y_{1i} \mid D = 1] &= E[Y_{1i} \mid D = 0] \end{aligned}$$

$$\underbrace{E[y_i | d_i = 1] - E[y_i | d_i = 0]}_{\text{SDO}} = \underbrace{E[Y_{1i}] - E[Y_{0i}]}_{\text{Average Treatment Effect}} + \underbrace{E[Y_{0i} | D = 1] - E[Y_{0i} | D = 0]}_{\text{Selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous treatment effect bias}}$$

- If treatment is independent of potential outcomes, then:

$$E[Y_{0i} | D = 1] - E[Y_{0i} | D = 0] = 0$$

Random Assignment

- How does randomization eliminate heterogeneous treatment effect bias?
Rewrite ATT and ATU:

$$\begin{aligned}\text{ATT} &= E[Y_{1i} | D = 1] - E[Y_{0i} | D = 1] \\ \text{ATU} &= E[Y_{1i} | D = 0] - E[Y_{0i} | D = 0]\end{aligned}$$

- Under independence, rewrite $\text{ATT} - \text{ATU}$:

$$\begin{aligned}\text{ATT} - \text{ATU} &= E[Y_{1i} | D = 1] - E[Y_{0i} | D = 1] \\ &\quad - E[Y_{1i} | D = 0] + E[Y_{0i} | D = 0] = 0\end{aligned}$$

- Therefore, if treatment is independent of potential outcomes:

$$E[y_i | D_i = 1] - E[y_i | D_i = 0] = E[Y_{1i}] - E[Y_{0i}] \quad \Rightarrow \quad \text{SDO} = \text{ATE}$$

- Define the following regression:

$$y_i = \hat{\beta}_0 + \hat{\delta} D_i + \hat{\epsilon}_i$$

- In ECON312 you showed that the OLS estimator of δ in a sample is:

$$\hat{\delta} = \frac{1}{n_1} \sum_{i=1}^{n_1} y_i - \frac{1}{n_0} \sum_{i=1}^{n_0} y_i$$

- From this equation, we obtain:

$$\underset{n_0, n_1 \rightarrow \infty}{\text{plim}} \hat{\delta} = E[y_i \mid D_i = 1] - E[y_i \mid D_i = 0] \underbrace{=}_{(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i} ATE$$

Randomization

- We can also think about the independence assumption in the context of the population regression:

$$y_i = \underbrace{\beta_0}_{E[y_{0i}]} + \underbrace{\delta}_{E[y_{1i}-y_{0i}]} D_i + \underbrace{\epsilon_i}_{y_{0i}-E[y_{0i}]+(\delta_i-\delta)D_i}$$

if randomly assigned:

$$E[y_{0i}|D_i = 1] = E[y_{0i}|D_i = 0] = E[y_{0i}]$$

and additionally $E[\delta_i|D_i = 1] = E[\delta_i] = \delta$ (ATT = ATE), which together imply:

$$E[\epsilon_i|D_i = 0] = E[y_{0i}|D_i = 0] - E[y_{0i}] = 0$$

$$\begin{aligned} E[\epsilon_i|D_i = 1] &= (E[y_{0i}|D_i = 1] - E[y_{0i}]) \\ &\quad + (E[\delta_i|D_i = 1] - \delta) = 0 \end{aligned}$$

$$\underbrace{\Rightarrow}_{\text{LIE}} E[\epsilon_i] = 0$$

- So the MM/OLS estimator is a consistent estimator of the **ATE**

Limits of Independence

- Full independence $(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i$ requires **randomized** treatment assignment
- In most observational data, treatment is **not** randomly assigned:
 - ▶ Students who attend university self-select based on ability and family background
 - ▶ Individuals who buy health insurance differ systematically from those who do not
 - ▶ Firms that adopt a policy differ systematically from those that do not
- Key question: can we make treatment “as good as random” *after* conditioning on observed characteristics $\mathbf{X}_i$?

- **Conditional Independence Assumption (CIA):**

$$(Y_{0i}, Y_{1i}) \perp\!\!\!\perp D_i \mid \mathbf{X}_i$$

- ▶ Also called *selection on observables*, *unconfoundedness*, or *ignorability*
- ▶ Requires **overlap**: $0 < P(D_i = 1 \mid \mathbf{X}_i) < 1$ (or a functional form assumption with within treatment group variation)

CIA: What It Buys You

- CIA implies that, **within cells defined by $\mathbf{X}_i$** , there is no selection bias, and $ATT=ATU$:

$$E[Y_{0i} \mid D_i = 1, \mathbf{X}_i] = E[Y_{0i} \mid D_i = 0, \mathbf{X}_i] = E[Y_{0i} \mid \mathbf{X}_i]$$

$$E[Y_{1i} \mid D_i = 1, \mathbf{X}_i] = E[Y_{1i} \mid D_i = 0, \mathbf{X}_i] = E[Y_{1i} \mid \mathbf{X}_i]$$

- Therefore, the **conditional** ATE is **identified** from observed data:

$$E[Y_i \mid D_i = 1, \mathbf{X}_i] - E[Y_i \mid D_i = 0, \mathbf{X}_i] = E[Y_{1i} - Y_{0i} \mid \mathbf{X}_i]$$

- **Importantly:** CIA does *not* say treatment is random — it says that *all* variables that jointly determine D_i and potential outcomes are **observed** in $\mathbf{X}_i$
- If **unobserved** confounders remain, CIA fails $\Rightarrow$ need IV, DiD, RD, ...

ATT, ATU, and ATE

- Three distinct parameters of interest, differing in the **population** over which we average:

$$\text{ATE} = \int E[Y_{1i} - Y_{0i} \mid \mathbf{X}_i = \mathbf{x}] \cdot f(\mathbf{x}) d\mathbf{x}$$

$$\text{ATT} = \int E[Y_{1i} - Y_{0i} \mid \mathbf{X}_i = \mathbf{x}] \cdot f(\mathbf{x} \mid D_i = 1) d\mathbf{x}$$

$$\text{ATU} = \int E[Y_{1i} - Y_{0i} \mid \mathbf{X}_i = \mathbf{x}] \cdot f(\mathbf{x} \mid D_i = 0) d\mathbf{x}$$

- $\text{ATT} \neq \text{ATU}$ because they **integrate over different distributions of $\mathbf{X}_i$** (unless the treatment is homogeneous)
- ATE is a **weighted average** of ATT and ATU by the Law of Iterated Expectations:

$$\text{ATE} = P(D_i = 1) \cdot \text{ATT} + P(D_i = 0) \cdot \text{ATU}$$

- (Note, under **random assignment**: $f(\mathbf{x} \mid D_i = 1) = f(\mathbf{x} \mid D_i = 0) = f(\mathbf{x})$, so $\text{ATT} = \text{ATU} = \text{ATE}$)

OLS with a binary Control: What Does Regression Identify?

- Under CIA, $y_i = \beta_0 + \delta^R D_i + \beta_1 A_i + u_i$
- With **heterogeneous** treatment effects $\delta_{A_i} \equiv E[y_{1i} - y_{0i} \mid A_i]$, then:

$$\delta^R = \frac{E[\text{Var}(D_i | A_i) \delta_{A_i}]}{E[\text{Var}(D_i | A_i)]}$$

- If $\delta_{A_i} = \delta$ for all A_i , then $\delta^R = \delta = ATE$
- On the other hand, when effect is heterogeneous, the weighting is different
- Despite the less straightforward weighting, the estimate is considered causal as it is a weighted average of causal effects
 - ▶ Differences between ATE and δ^R are small when the heterogeneity is small
- Linear regression still a powerful tool, in particular, when the covariate is continuous as the linearity does not require overlap, but that the covariate varies within treatment
 - ▶ Nevertheless, the in the last few decades matching estimators that directly estimate *ATT* and *ATE* has become more popular due to its direct interpretability (more of this later, time permitting)

Omitted Variable Bias

- Suppose the **true** population model includes a confounder X_i :

$$y_i = \beta_0 + \delta D_i + \beta_1 X_i + u_i, \quad E[u_i \mid D_i, X_i] = 0$$

- If we **omit** X_i and estimate $y_i = \hat{\beta}_0 + \hat{\delta} D_i + \hat{\epsilon}_i$, and, $\text{Cov}(X_i, D_i) \neq 0$, then $\text{Cov}(\epsilon_i, D_i) \neq 0$ and OLS is **inconsistent**:

$$\text{plim}_{n \rightarrow \infty} \hat{\delta} = \delta + \beta_1 \underbrace{\frac{\text{Cov}(X_i, D_i)}{\text{Var}(D_i)}}_{\equiv \gamma_1}$$

where γ_1 is the slope from the auxiliary regression $X_i = \gamma_0 + \gamma_1 D_i + v_i$

- The bias $\beta_1 \gamma_1$ has sign determined by: (i) whether X_i raises y_i ($\beta_1 \geq 0$) and (ii) whether treated units have higher X_i ($\gamma_1 \geq 0$)
- Solution:** control for X_i directly $\Rightarrow$ under CIA, OLS on the long regression identifies δ

Bad Controls

- As is the case with omitted variable bias, adding **bad** controls in the regression leads to inconsistent causal estimates
- A **bad control** is usually a variable that is itself an *outcome* of the treatment, and adding it to the regression results in the following
 1. It **blocks part of the causal channel** you are trying to measure, which changes the interpretation of the causal estimate
 2. If the control does not satisfy the CIA, it introduces selection bias even if CIA or unconditional independence holds for the specification without the control
- When 2. happens, MHE refers to this control as a **bad control**
 - ▶ **Bottom line:** when uncertain whether a control is good or bad, ask — “*Could the treatment change this variable?*” If yes, either think carefully about what the introduction of this variable does to the regression, or leave it out.

Bad Controls: The Formal Argument

- Suppose Y_i is wages, D_i is university degree (treatment), and W_i is white collar occupation.
 - ▶ Assume that $Y_{D_i=0,i}, Y_{D_i=1,i} \perp\!\!\!\perp D_i$ so that the 'short' regression, identifies a causal effect:

$$Y_i = \beta_0 + \delta^{ATE} D_i + \epsilon \quad , \quad E[\epsilon_i | D_i] = E[\epsilon_i]$$

- ▶ Assume that $Y_{W_i=0,i}, Y_{W_i=1,i} \not\perp\!\!\!\perp W_i | D_i$ so that the 'long' regression, introduces selection bias:

$$Y_i = \beta_0 + \delta^R D_i + \beta_1 W_i + u_i \quad , \quad E[u_i | D_i, W_i] \neq E[u_i]$$